CITY HOSPITAL

Network Topology Design

Complex Computing Problem | CT-376 | Spring 2026

Submitted To Sir Abdul Karim Kazi

Department of Computer Science & Information Technology

Group-31

Member 1 Aayan Altaf CT-23075
Member 2 Muhammad Abdullah CT-23077
Member 3 Muhammad Ashad Khan CT-23084
Member 4 Zubair Ullah Khan CT-23087

Simulation Tool: Huawei eNSP Simulator

Assessment Rubrics

Course Code: CT-376 **Course Title:** Computer Communication Networks **Total Marks:** 08

Criteria	Excellent (2)	Average / Good (1)	Poor (0)
1. Understanding the Problem	Demonstrates a comprehensive understanding of the problem statement; clearly identifies all underlying issues and required functionalities.	Adequately understands the problem; identifies most underlying issues and functionalities.	Fails to adequately define the problem or identify underlying issues.
2. Research & Design	Topology reflects thorough research; all necessary information and configurations present to solve the problem effectively.	Research is mediocre; solution is partial or lacks depth in implementation.	No apparent research or viable solution provided.
3. Documentation & Report	Report is professional, thoroughly organized, and provides all necessary information for easy interpretation of the solution.	Information is disorganized or mediocre; may not fully support the solution presented.	No report provided.
4. Labeling & Presentation	Simulation is meticulously labeled — port numbers, IP addresses, VLAN tags — to facilitate easy understanding.	Labels are incomplete or inconsistent regarding selected functionalities.	Topology is not labeled.

Obtained Marks: _____ Teacher's Signature: _____

1. Introduction

1.1 Project Overview

This report documents the complete design, IP addressing scheme, device configuration, and verification of a complex enterprise-grade network topology built for City Hospital. The network is implemented and simulated using the Huawei eNSP simulator and fulfills the Complex Computing Problem (CCP) requirements for the course Computer Communication Networks (CT-376).

The topology spans five hospital departments — Patient Registration, Administration, Pharmacy, Emergency & ICU, and Radiology — each with its own internal network design and primary networking technology. All departments are interconnected through a central hub router (HospCore-R1) using a hub-and-spoke backbone with RIP v2 dynamic routing. A dedicated FTP server provides centralized file services to all departments.

1.2 Motivation & Domain

A hospital was chosen as the project domain because it naturally demands a wide range of networking technologies in a single deployment. Admission terminals need DHCP for dynamic IP management across shifts. ICU monitoring systems must be isolated from general ward access using ACLs. Administration requires VLAN separation between board members and general staff. Radiology needs predictable, controlled data paths for imaging files via static routing. The hospital context makes every technology choice meaningful and justifiable rather than academic.

1.3 Technologies Implemented

- VLSM (Variable Length Subnet Masking) — efficient IP addressing across all departments
- DHCP — dynamic IP assignment for Patient Registration terminals
- VLANs — logical network segmentation in Administration, Pharmacy, and Emergency
- Router on a Stick — inter-VLAN routing via sub-interfaces on department routers
- STP (Spanning Tree Protocol) — loop prevention in Administration switch triangle
- Eth-Trunk (Link Aggregation) — redundant, high-bandwidth link in Administration
- Static Routing — controlled path management in Radiology router triangle
- RIP v2 (Routing Information Protocol) — dynamic inter-department backbone routing
- ACL (Access Control List) — ICU protection from General Ward traffic
- FTP (File Transfer Protocol) — centralised hospital document file service
- Telnet — remote management of all department routers from HospCore-R1

2. Network Topology

2.1 Topology Design

The overall topology uses a hub-and-spoke architecture at the backbone level. HospCore-R1 is the central hub, connecting to six departmental routers via dedicated /30 point-to-point links. Each departmental router independently manages its internal LAN using the networking technology most appropriate for that department's requirements.

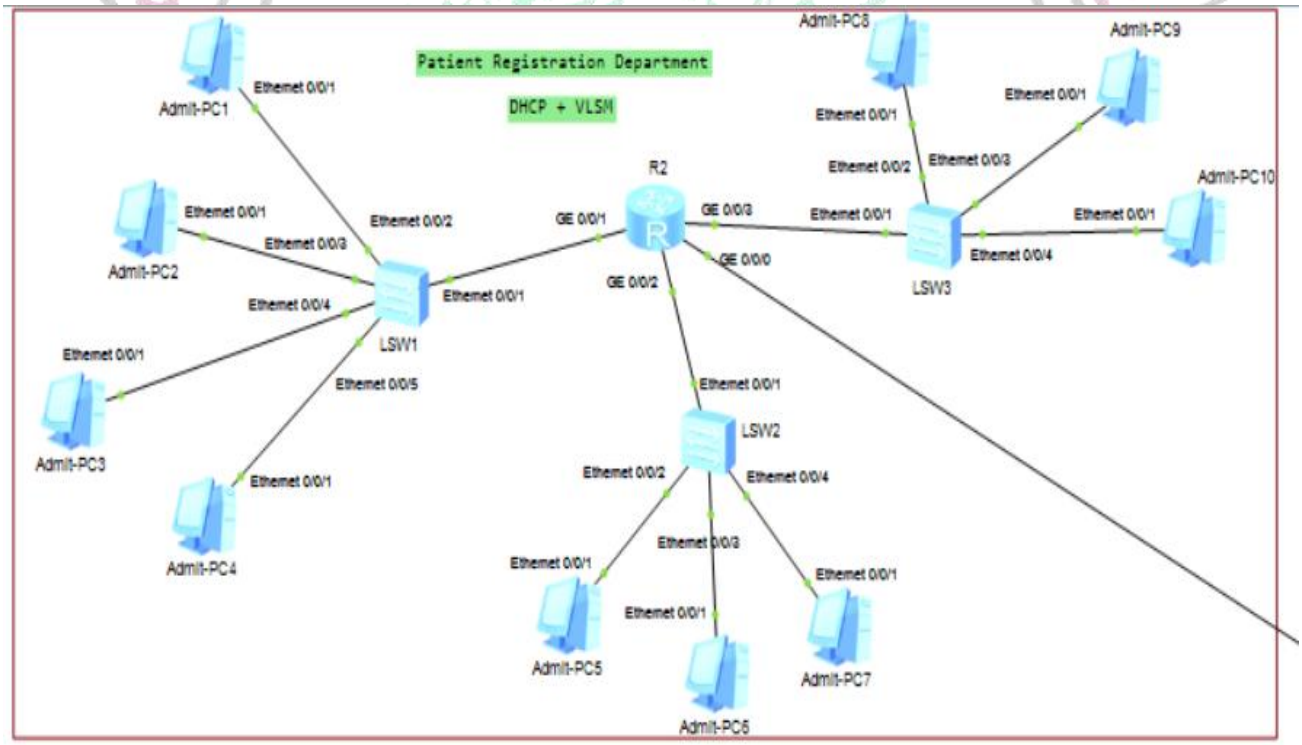
The backbone links all use the 10.0.x.0/30 address space, keeping backbone addressing completely separate from departmental addressing. RIP v2 with auto-summary disabled runs on all backbone routers to dynamically distribute departmental subnet information across the entire network.

2.2 Department Summaries

Patient Registration Department

Router: PatReg-R2 | Technology: DHCP + VLSM | Switches: PatReg-SW1, SW2, SW3

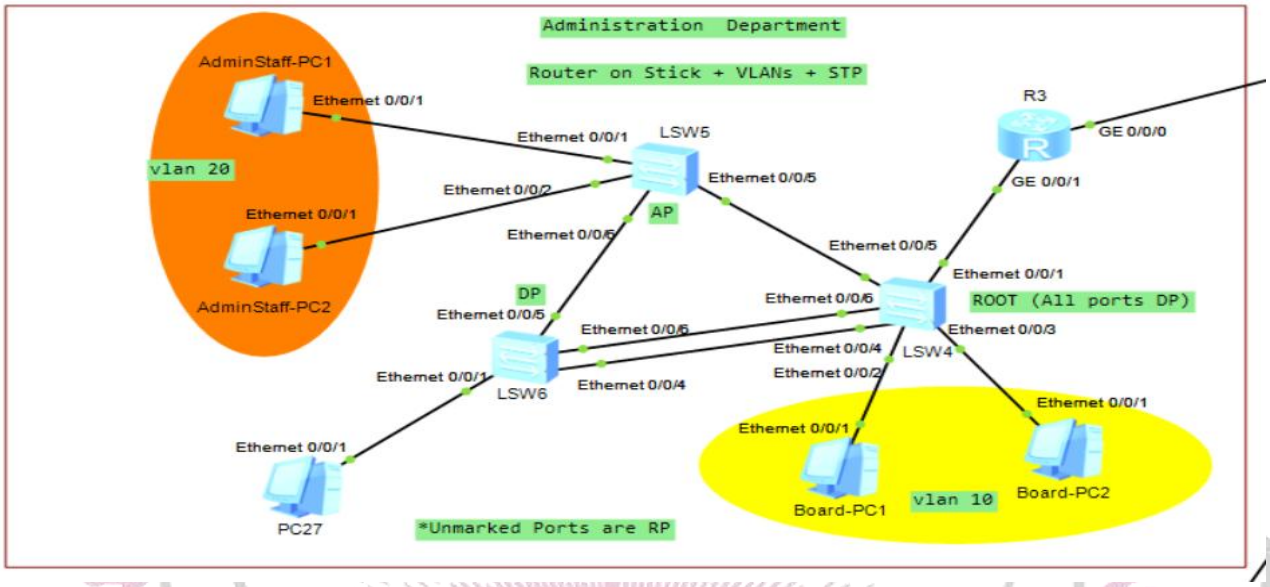
Three floor switches serve a total of 10 admission terminals across different registration counters. PatReg-R2 acts as the DHCP server with one pool per switch, automatically assigning /29 subnet addresses. Terminals change between shifts so manual IP management would be impractical.



Administration Department

Router: AdminGW-R3 | Technologies: VLANs, Router on a Stick, STP, Eth-Trunk | Switches: Admin-SW4, SW5, SW6

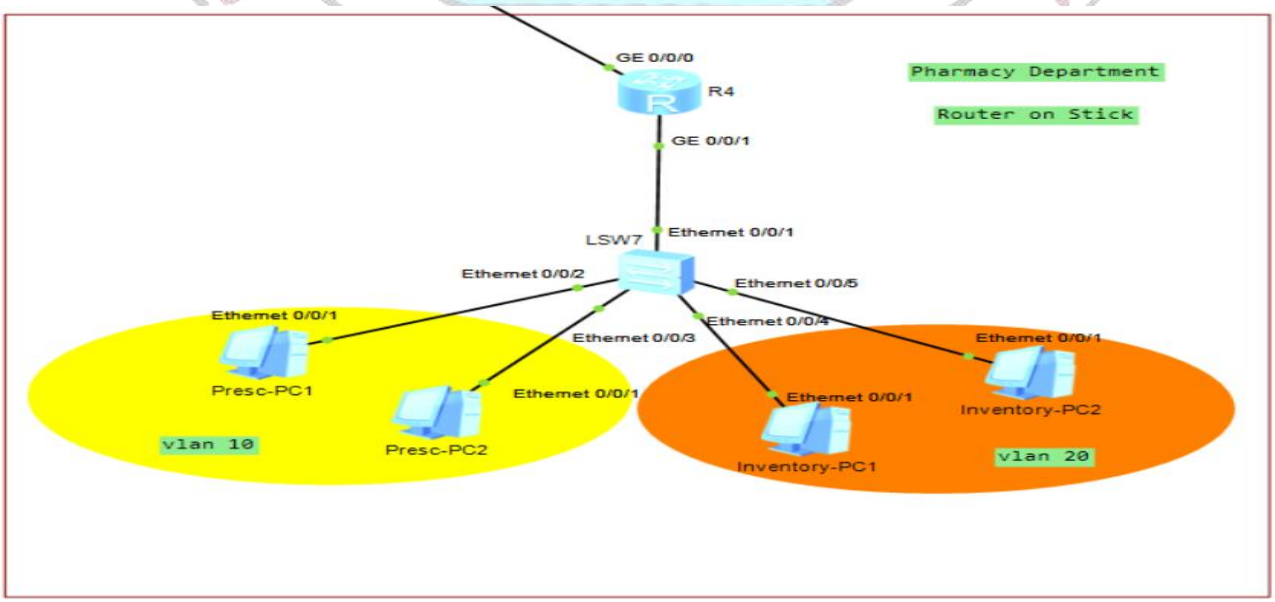
The administration network separates Hospital Board members (VLAN 10) from general Administrative Staff (VLAN 20) on a three-switch infrastructure. AdminGW-R3 performs inter-VLAN routing. Admin-SW4 is the STP root bridge and maintains an Eth-Trunk to Admin-SW6 for redundancy. STP prevents broadcast storms in the triangle formed by all three switches.



Pharmacy Department

Router: PharmaGW-R4 | Technology: VLANs, Router on a Stick | Switch: Pharma-SW7

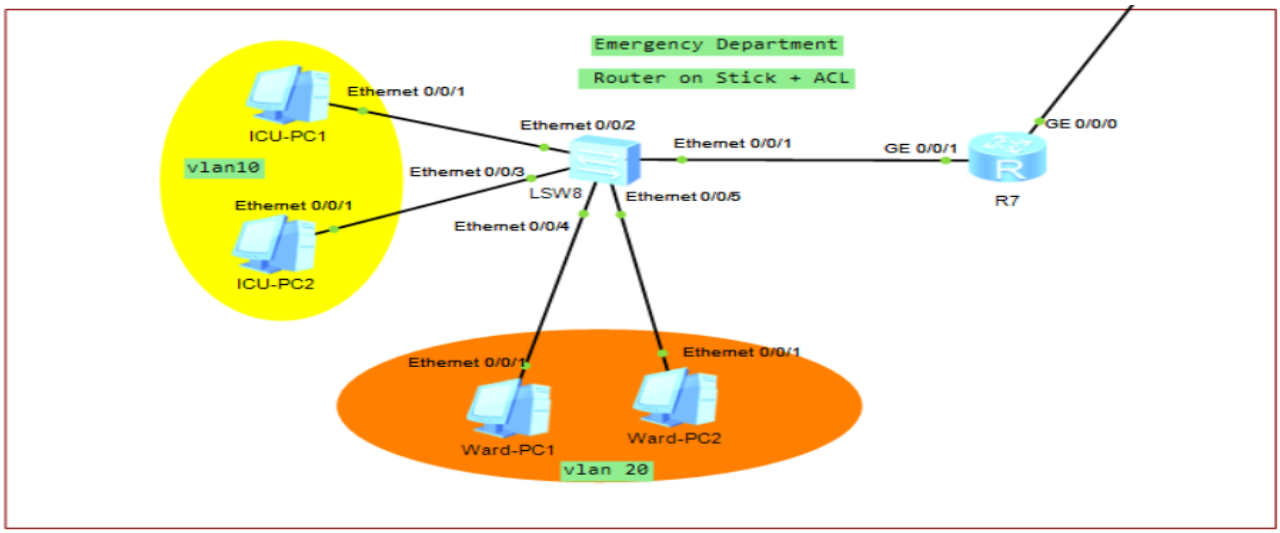
A single switch separates Prescription Counter terminals (VLAN 10) from Inventory Management systems (VLAN 20). PharmaGW-R4 provides inter-VLAN routing via sub-interfaces, allowing both systems to communicate through the router while remaining Layer 2 isolated.



Emergency & ICU Department

Router: EmergGW-R7 | Technologies: VLANs, Router on a Stick, ACL | Switch: Emerg-SW8

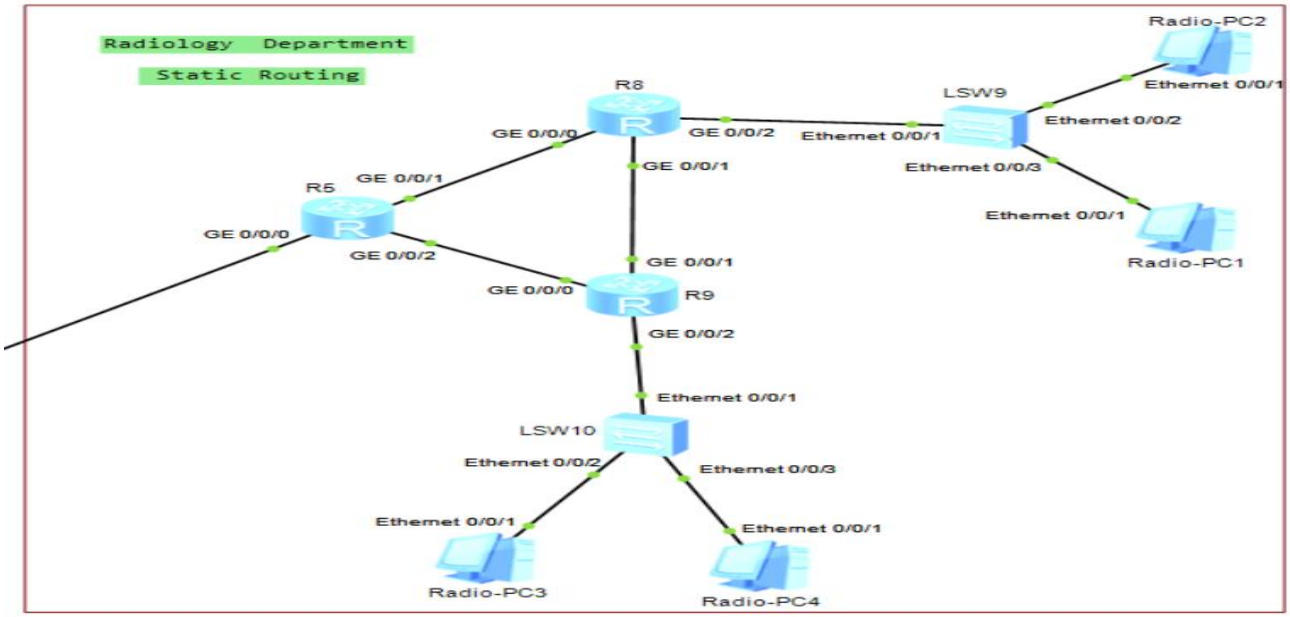
ICU monitoring stations (VLAN 10) and General Ward PCs (VLAN 20) share the same physical switch but are logically separated by VLANs. ACL 2000 on EmergGW-R7 denies any traffic from the General Ward subnet from reaching the ICU sub-interface outbound, protecting critical patient monitoring systems.



Radiology Department

Routers: RadioGW-R5, RadioR8, RadioR9 | Technology: Static Routing | Switches: Radio-SW9, SW10

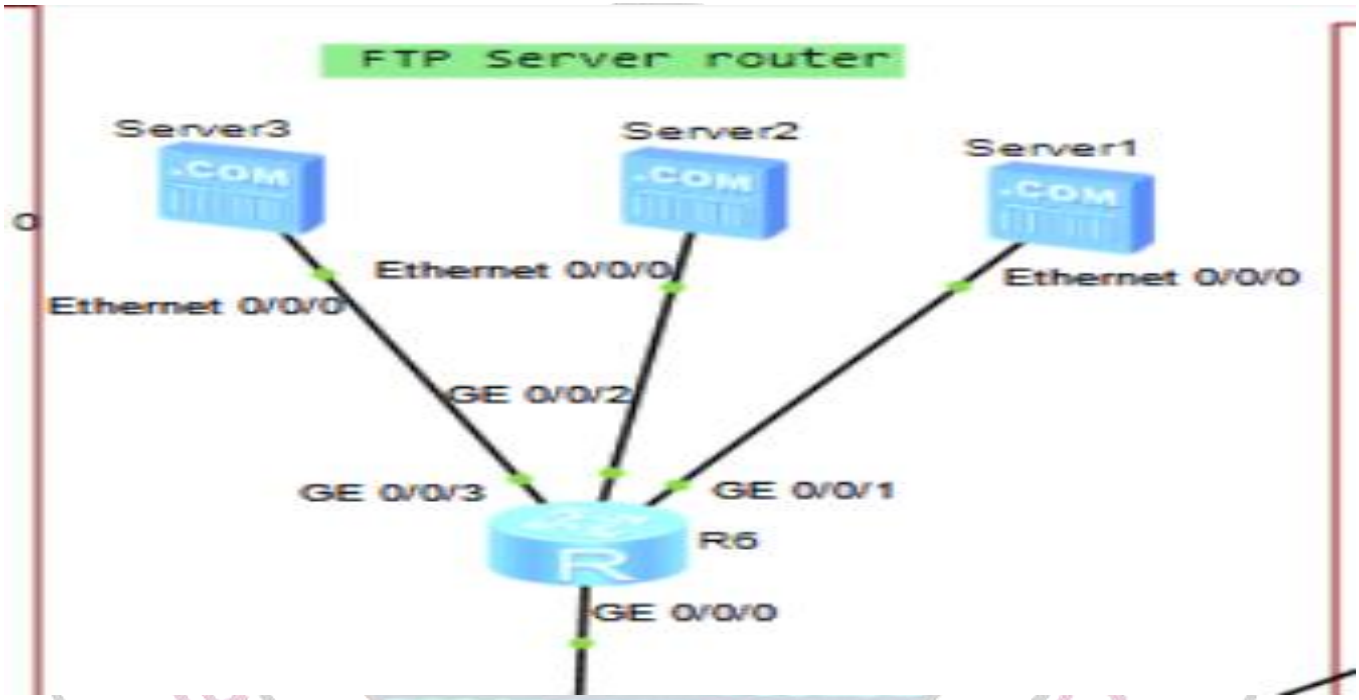
Three routers form a triangle with manually configured static routes. Each path is explicitly defined with dual next-hop entries for redundancy. RadioR8 and RadioR9 each serve a LAN switch with radiologist workstations. Static routing ensures DICOM imaging data always follows predictable, administrator-controlled paths.



FTP Servers

Router: FTPSrv-R6 | Technology: FTP | Devices: Server1 (Primary), Server2 (Backup), Server3 (Archive)

FTPSrv-R6 acts as a dedicated medical records gateway managing three FTP servers for redundancy and data segregation. Server1 (1.1.1.2) handles active patient records and report templates. Server2 (1.1.2.2) stores departmental backups. Server3 (1.1.3.2) archives discharged patient history. All three servers are reachable from every department router via RIP, and FTPSrv-R6 connects to each server on separate /30 subnets through GE0/0/1, GE0/0/2, and GE0/0/3 respectively.




```
<FTPSrv-R6>ftp 1.1.2.2
Trying 1.1.2.2 ...
Press CTRL+K to abort
Connected to 1.1.2.2.
220 FtpServerTry FtpD for free
User(1.1.2.2:(none)):
331 Password required for  .
Enter password:
230 User  logged in , proceed

[ftp]ls
200 Port command okay.
```

3. IP Addressing Scheme

3.1 Backbone Point-to-Point Links

All backbone links between HospCore-R1 and departmental routers use /30 subnets from the 10.0.0.0/8 block. /30 gives exactly 2 usable host addresses — one per router end — eliminating address waste on transit links.

Link	Network	R1 Side	Dept Router Side	Mask
R1 ↔ PatReg-R2	10.0.1.0/30	10.0.1.1	10.0.1.2	255.255.255.252
R1 ↔ AdminGW-R3	10.0.2.0/30	10.0.2.1	10.0.2.2	255.255.255.252
R1 ↔ PharmaGW-R4	10.0.3.0/30	10.0.3.1	10.0.3.2	255.255.255.252
R1 ↔ RadioGW-R5	10.0.4.0/30	10.0.4.1	10.0.4.2	255.255.255.252
R1 ↔ FTPSrv-R6	10.0.5.0/30	10.0.5.1	10.0.5.2	255.255.255.252
R1 ↔ EmergGW-R7	10.0.6.0/30	10.0.6.1	10.0.6.2	255.255.255.252
R5 ↔ RadioR8	172.16.1.0/30	172.16.1.1	172.16.1.2	255.255.255.252
R5 ↔ RadioR9	172.16.2.0/30	172.16.2.1	172.16.2.2	255.255.255.252
RadioR8 ↔ RadioR9	172.16.3.0/30	172.16.3.1	172.16.3.2	255.255.255.252

3.2 VLSM LAN Subnets

All LAN-facing subnets are sized using VLSM to match the number of hosts exactly. Every department LAN uses /29 subnets (6 usable hosts each), calculated from a base /24 class-C network assigned per department.

Subnet	Prefix	Network Address	Usable Range	Broadcast	Gateway	Dept / Purpose
/29	192.168.1.0/29	192.168.1.0	192.168.1.2 – .6	192.168.1.7	192.168.1.1	PatReg SW1 (4 PCs)
/29	192.168.1.8/29	192.168.1.8	192.168.1.10 – .14	192.168.1.15	192.168.1.9	PatReg SW2 (3 PCs)
/29	192.168.1.16/29	192.168.1.16	192.168.1.18 – .22	192.168.1.23	192.168.1.17	PatReg SW3 (3 PCs)
/29	192.168.10.0/29	192.168.10.0	192.168.10.2 – .6	192.168.10.7	192.168.10.1	Admin VLAN 10 — Board
/29	192.168.10.8/29	192.168.10.8	192.168.10.10 – .14	192.168.10.15	192.168.10.9	Admin VLAN 20 — Staff
/29	192.168.20.0/29	192.168.20.0	192.168.20.2 – .6	192.168.20.7	192.168.20.1	Pharma VLAN 10 — Prescription
/29	192.168.20.8/29	192.168.20.8	192.168.20.10 – .14	192.168.20.15	192.168.20.9	Pharma VLAN 20 — Inventory
/29	192.168.30.0/29	192.168.30.0	192.168.30.2 – .6	192.168.30.7	192.168.30.1	Emergency VLAN 10 — ICU
/29	192.168.30.8/29	192.168.30.8	192.168.30.10 – .14	192.168.30.15	192.168.30.9	Emergency VLAN 20 — Ward
/29	172.16.10.0/29	172.16.10.0	172.16.10.2 – .6	172.16.10.7	172.16.10.1	RadioR8 LAN (Radiology)
/29	172.16.11.0/29	172.16.11.0	172.16.11.2 – .6	172.16.11.7	172.16.11.1	RadioR9 LAN (Radiology)
/30	1.1.1.0/30	1.1.1.0	1.1.1.1 – 1.1.1.2	1.1.1.3	1.1.1.1	FTP Server link

Subnet	Prefix	Network Address	Usable Range	Broadcast	Gateway	Dept / Purpose
/30	1.1.2.0/30	1.1.2.0	1.1.2.1 – 1.1.2.2	1.1.2.3	1.1.2.1	FTP Server2 link (Backup)
/30	1.1.3.0/30	1.1.3.0	1.1.3.1 – 1.1.3.2	1.1.3.3	1.1.3.1	FTP Server3 link (Archive)

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4. Device Inventory

4.1 Routers

Device Name	Model	Department	Primary Technologies
HospCore-R1	Router	Core / Backbone	RIP v2, Telnet server, 4x GE, 2x Eth backbone interfaces
PatReg-R2	Router	Patient Registration	DHCP server (3 pools), VLSM /29, RIP v2
AdminGW-R3	Router	Administration	Router on Stick (sub-intf), RIP v2, Telnet
PharmaGW-R4	Router	Pharmacy	Router on Stick (sub-intf), RIP v2, Telnet
RadioGW-R5	Router	Radiology	Static routing (dual paths), RIP v2 to backbone
FTPSrv-R6	Router	FTP Server	FTP gateway — 3 servers (Primary/Backup/Archive), RIP v2
EmergGW-R7	Router	Emergency & ICU	Router on Stick, ACL 2000 (deny ward→ICU), RIP v2
RadioR8	Router	Radiology	Static routing, LAN gateway for Radio-SW9
RadioR9	Router	Radiology	Static routing, LAN gateway for Radio-SW10

4.2 Switches

Device Name	Model	Department	Role & Notes
PatReg-SW1	S3700	Patient Registration	4 access ports for Admit-PC1–4; uplink GE to R2 GE0/0/1
PatReg-SW2	S3700	Patient Registration	3 access ports for Admit-PC5–7; uplink GE to R2 GE0/0/2
PatReg-SW3	S3700	Patient Registration	3 access ports for Admit-PC8–10; uplink GE to R2 GE0/0/3
Admin-SW4	S3700	Administration	STP Root Bridge (priority 4096); Eth-Trunk to SW6; trunk uplink to R3
Admin-SW5	S3700	Administration	Non-root STP; VLAN 20 access for staff PCs; one port DISCARDING
Admin-SW6	S3700	Administration	Eth-Trunk partner with SW4 (Eth0/0/4 & Eth0/0/6); trunk to SW5
Pharma-SW7	S3700	Pharmacy	VLAN 10 access (Eth0/0/2–3) + VLAN 20 access (Eth0/0/4–5); trunk to R4
Emerg-SW8	S3700	Emergency & ICU	VLAN 10 access (Eth0/0/2–3 ICU) + VLAN 20 access (Eth0/0/4–5 Ward); trunk to R7

Device Name	Model	Department	Role & Notes
Radio-SW9	S3700	Radiology	LAN switch connected to RadioR8 GE0/0/2; hosts Radio-PC1 and PC2
Radio-SW10	S3700	Radiology	LAN switch connected to RadioR9 GE0/0/2; hosts Radio-PC3 and PC4

5. Complete Endpoint IP Assignment Table

The table below lists every end device in the topology with its complete IP configuration. Devices in Patient Registration use DHCP; all other devices are statically configured.

Device	IP Address	Subnet Mask	Default Gateway	VLAN	Method	Department
Admit-PC1	192.168.1.2	255.255.255.248	192.168.1.1	—	DHCP	Patient Registration
Admit-PC2	192.168.1.3	255.255.255.248	192.168.1.1	—	DHCP	Patient Registration
Admit-PC3	192.168.1.4	255.255.255.248	192.168.1.1	—	DHCP	Patient Registration
Admit-PC4	192.168.1.5	255.255.255.248	192.168.1.1	—	DHCP	Patient Registration
Admit-PC5	192.168.1.10	255.255.255.248	192.168.1.9	—	DHCP	Patient Registration
Admit-PC6	192.168.1.11	255.255.255.248	192.168.1.9	—	DHCP	Patient Registration
Admit-PC7	192.168.1.12	255.255.255.248	192.168.1.9	—	DHCP	Patient Registration
Admit-PC8	192.168.1.18	255.255.255.248	192.168.1.17	—	DHCP	Patient Registration
Admit-PC9	192.168.1.19	255.255.255.248	192.168.1.17	—	DHCP	Patient Registration
Admit-PC10	192.168.1.20	255.255.255.248	192.168.1.17	—	DHCP	Patient Registration
Board-PC1	192.168.10.2	255.255.255.248	192.168.10.1	10	Static	Administration
Board-PC2	192.168.10.3	255.255.255.248	192.168.10.1	10	Static	Administration
AdminStaff-PC1	192.168.10.10	255.255.255.248	192.168.10.9	20	Static	Administration
AdminStaff-PC2	192.168.10.11	255.255.255.248	192.168.10.9	20	Static	Administration
Presc-PC1	192.168.20.2	255.255.255.248	192.168.20.1	10	Static	Pharmacy
Presc-PC2	192.168.20.3	255.255.255.248	192.168.20.1	10	Static	Pharmacy
Inventory-PC1	192.168.20.10	255.255.255.248	192.168.20.9	20	Static	Pharmacy
Inventory-PC2	192.168.20.11	255.255.255.248	192.168.20.9	20	Static	Pharmacy

Device	IP Address	Subnet Mask	Default Gateway	VLAN	Method	Department
ICU-PC1	192.168.30.2	255.255.255.248	192.168.30.1	10	Static	Emergency & ICU
ICU-PC2	192.168.30.3	255.255.255.248	192.168.30.1	10	Static	Emergency & ICU
Ward-PC1	192.168.30.10	255.255.255.248	192.168.30.9	20	Static	Emergency & ICU
Ward-PC2	192.168.30.11	255.255.255.248	192.168.30.9	20	Static	Emergency & ICU
Radio-PC1	172.16.10.2	255.255.255.248	172.16.10.1	—	Static	Radiology
Radio-PC2	172.16.10.3	255.255.255.248	172.16.10.1	—	Static	Radiology
Radio-PC3	172.16.11.2	255.255.255.248	172.16.11.1	—	Static	Radiology
Radio-PC4	172.16.11.3	255.255.255.248	172.16.11.1	—	Static	Radiology
Server1 (Primary)	1.1.1.2	255.255.255.252	1.1.1.1	—	Static	FTP Server
Server2 (Backup)	1.1.2.2	255.255.255.252	1.1.2.1	—	Static	FTP Server
Server3 (Archive)	1.1.3.2	255.255.255.252	1.1.3.1	—	Static	FTP Server

6. Physical Connection Table

All connections use standard Copper Ethernet cables in eNSP. The two cables marked as Eth-Trunk members form a logical aggregated link.

From Device	From Port	To Device	To Port	Link Type / Purpose
HospCore-R1	GE0/0/0	PatReg-R2	GE0/0/0	Backbone /30 — R1 to PatReg
HospCore-R1	GE0/0/1	AdminGW-R3	GE0/0/0	Backbone /30 — R1 to Admin
HospCore-R1	GE0/0/2	PharmaGW-R4	GE0/0/0	Backbone /30 — R1 to Pharma
HospCore-R1	GE0/0/3	RadioGW-R5	Eth0/0/0	Backbone /30 — R1 to Radiology
HospCore-R1	GE0/0/4	FTPSrv-R6	GE0/0/0	Backbone /30 — R1 to FTP
HospCore-R1	GE0/0/5	EmergGW-R7	Eth0/0/1	Backbone /30 — R1 to Emergency
PatReg-R2	GE0/0/1	PatReg-SW1	Eth0/0/1	DHCP Pool 1 — 4 PC floor
PatReg-R2	GE0/0/2	PatReg-SW2	Eth0/0/1	DHCP Pool 2 — 3 PC floor
PatReg-R2	GE0/0/3	PatReg-SW3	Eth0/0/1	DHCP Pool 3 — 3 PC floor
PatReg-SW1	Eth0/0/2	Admit-PC1	Eth0/0/1	Access — DHCP client
PatReg-SW1	Eth0/0/3	Admit-PC2	Eth0/0/1	Access — DHCP client
PatReg-SW1	Eth0/0/4	Admit-PC3	Eth0/0/1	Access — DHCP client
PatReg-SW1	Eth0/0/5	Admit-PC4	Eth0/0/1	Access — DHCP client
PatReg-SW2	Eth0/0/2	Admit-PC5	Eth0/0/1	Access — DHCP client
PatReg-SW2	Eth0/0/3	Admit-PC6	Eth0/0/1	Access — DHCP client
PatReg-SW2	Eth0/0/4	Admit-PC7	Eth0/0/1	Access — DHCP client
PatReg-SW3	Eth0/0/2	Admit-PC8	Eth0/0/1	Access — DHCP client
PatReg-SW3	Eth0/0/3	Admit-PC9	Eth0/0/1	Access — DHCP client
PatReg-SW3	Eth0/0/4	Admit-PC10	Eth0/0/1	Access — DHCP client
AdminGW-R3	GE0/0/1	Admin-SW4	Eth0/0/1	Trunk — carries VLAN 10 & 20
Admin-SW4	Eth0/0/2	Board-PC1	Eth0/0/1	Access — VLAN 10
Admin-SW4	Eth0/0/3	Board-PC2	Eth0/0/1	Access — VLAN 10
Admin-SW4	Eth0/0/4	Admin-SW6	Eth0/0/4	Eth-Trunk link 1 (aggregated)
Admin-SW4	Eth0/0/6	Admin-SW6	Eth0/0/6	Eth-Trunk link 2 (aggregated)
Admin-SW4	Eth0/0/5	Admin-SW5	Eth0/0/5	Trunk — STP active path

From Device	From Port	To Device	To Port	Link Type / Purpose
Admin-SW5	Eth0/0/1	AdminStaff-PC1	Eth0/0/1	Access — VLAN 20
Admin-SW5	Eth0/0/2	AdminStaff-PC2	Eth0/0/1	Access — VLAN 20
Admin-SW5	Eth0/0/6	Admin-SW6	Eth0/0/5	Trunk — STP blocks this port
PharmaGW-R4	GE0/0/1	Pharma-SW7	Eth0/0/1	Trunk — carries VLAN 10 & 20
Pharma-SW7	Eth0/0/2	Presc-PC1	Eth0/0/1	Access — VLAN 10
Pharma-SW7	Eth0/0/3	Presc-PC2	Eth0/0/1	Access — VLAN 10
Pharma-SW7	Eth0/0/4	Inventory-PC1	Eth0/0/1	Access — VLAN 20
Pharma-SW7	Eth0/0/5	Inventory-PC2	Eth0/0/1	Access — VLAN 20
EmergGW-R7	GE0/0/1	Emerg-SW8	Eth0/0/1	Trunk — carries VLAN 10 & 20
Emerg-SW8	Eth0/0/2	ICU-PC1	Eth0/0/1	Access — VLAN 10 (ICU)
Emerg-SW8	Eth0/0/3	ICU-PC2	Eth0/0/1	Access — VLAN 10 (ICU)
Emerg-SW8	Eth0/0/4	Ward-PC1	Eth0/0/1	Access — VLAN 20 (Ward)
Emerg-SW8	Eth0/0/5	Ward-PC2	Eth0/0/1	Access — VLAN 20 (Ward)
RadioGW-R5	GE0/0/1	RadioR8	GE0/0/0	Static link — /30 transit
RadioGW-R5	GE0/0/2	RadioR9	GE0/0/0	Static link — /30 transit
RadioR8	GE0/0/1	RadioR9	GE0/0/1	Static link — R8-R9 triangle leg
RadioR8	GE0/0/2	Radio-SW9	Eth0/0/1	LAN uplink — Radiology wing A
RadioR9	GE0/0/2	Radio-SW10	Eth0/0/1	LAN uplink — Radiology wing B
Radio-SW9	Eth0/0/2	Radio-PC1	Eth0/0/1	Access — Radiologist workstation
Radio-SW9	Eth0/0/3	Radio-PC2	Eth0/0/1	Access — Radiologist workstation
Radio-SW10	Eth0/0/2	Radio-PC3	Eth0/0/1	Access — Radiologist workstation
Radio-SW10	Eth0/0/3	Radio-PC4	Eth0/0/1	Access — Radiologist workstation
FTPSrv-R6	GE0/0/1	MedFTP-Server	Eth0/0/0	Primary FTP server link — /30
FTPSrv-R6	GE0/0/2	Server2	Eth0/0/0	Backup FTP server link — /30
FTPSrv-R6	GE0/0/3	Server3	Eth0/0/0	Archive FTP server link — /30

7. Technology Configuration & Justification

7.1 VLSM — Variable Length Subnet Masking

VLSM is applied across all departments so that the size of each subnet exactly matches the number of hosts required, rather than allocating a uniform /24 to every network segment. Backbone point-to-point links receive /30 subnets (2 usable hosts), and all LAN segments receive /29 subnets (6 usable hosts).

VLSM Justification:

Assigning a /24 per department segment would allocate 254 usable addresses to every floor switch, wasting 248–250 addresses per subnet. With VLSM, the same 11 LAN segments consume only 88 usable addresses in total. This is especially important in a hospital environment where the IT team must manage hundreds of network segments across multiple buildings, and address space conservation directly reduces routing table size and administrative overhead.

7.2 DHCP — Dynamic Host Configuration Protocol

PatReg-R2 is configured as the DHCP server for the Patient Registration department. Three separate IP pools are created — one for each floor switch — and the DHCP select global command is applied on each LAN-facing interface so the router assigns addresses from the pool matching that interface's subnet.

```
dhcp enable

ip pool PatReg-Pool1
  gateway-list 192.168.1.1
  network 192.168.1.0 mask 255.255.255.248

ip pool PatReg-Pool2
  gateway-list 192.168.1.9
  network 192.168.1.8 mask 255.255.255.248

ip pool PatReg-Pool3
  gateway-list 192.168.1.17
  network 192.168.1.16 mask 255.255.255.248

interface GigabitEthernet0/0/1
  ip address 192.168.1.1 255.255.255.248
  dhcp select global

interface GigabitEthernet0/0/2
  ip address 192.168.1.9 255.255.255.248
  dhcp select global

interface GigabitEthernet0/0/3
  ip address 192.168.1.17 255.255.255.248
  dhcp select global
```

DHCP Justification:

Admission terminals are operated by shift workers who may use different machines each shift. Manually configuring static IPs on every terminal would require an IT administrator to visit and configure each device individually. DHCP eliminates this entirely — any terminal that powers on and

connects to a floor switch automatically receives a valid IP address, subnet mask, and gateway, making it immediately operational without any manual intervention.

7.3 VLANs — Virtual Local Area Networks

VLANs are deployed in Administration (VLAN 10: Board, VLAN 20: Admin Staff), Pharmacy (VLAN 10: Prescription, VLAN 20: Inventory), and Emergency (VLAN 10: ICU, VLAN 20: General Ward). Access ports assign end devices to their VLAN. Trunk ports carry both VLANs between switches and up to the router for inter-VLAN routing.

```
# Example: Pharma-SW7 VLAN configuration
vlan batch 10 20

interface Ethernet0/0/1
  port link-type trunk
  port trunk allow-pass vlan 10 20

interface Ethernet0/0/2
  port link-type access
  port default vlan 10

interface Ethernet0/0/3
  port link-type access
  port default vlan 10

interface Ethernet0/0/4
  port link-type access
  port default vlan 20

interface Ethernet0/0/5
  port link-type access
  port default vlan 20
```

VLAN Justification:

VLANs create logical broadcast domains within a shared physical switch. Hospital Board members' financial traffic and strategic documents are isolated from general administrative staff desktops at Layer 2, meaning a network sniffer on an admin staff machine cannot capture board-level traffic. Similarly, ICU patient monitoring traffic is isolated from general ward PCs. This improves both security and performance by shrinking broadcast domains.

7.4 Router on a Stick — Inter-VLAN Routing

Three department routers (AdminGW-R3, PharmaGW-R4, EmergGW-R7) use the Router on a Stick technique. A single physical GigabitEthernet interface connects the router to a trunk switch port. The router creates sub-interfaces for each VLAN, applies a dot1q termination VID tag, and assigns an IP address. The arp broadcast enable command ensures ARP resolution works correctly on sub-interfaces.

```
# Example: AdminGW-R3 sub-interface config
interface GigabitEthernet0/0/1.10
  dot1q termination vid 10
  ip address 192.168.10.1 255.255.255.248
  arp broadcast enable
```

```
interface GigabitEthernet0/0/1.20
  dot1q termination vid 20
  ip address 192.168.10.9 255.255.255.248
  arp broadcast enable
```

Router on a Stick Justification:

Without inter-VLAN routing, a Board member PC on VLAN 10 could never communicate with an Admin Staff PC on VLAN 20, even for legitimate tasks such as accessing shared hospital management systems. Router on a Stick allows this communication to flow through the router where policies can be applied, while keeping only one physical cable between the switch and router — saving switch ports and cabling costs.

7.5 STP — Spanning Tree Protocol

Admin-SW4, Admin-SW5, and Admin-SW6 are physically connected in a triangle for redundancy. Without STP, a broadcast frame entering the triangle would loop forever, flooding all bandwidth. STP is enabled on all three switches. Admin-SW4 is set as the root bridge by assigning it a lower priority of 4096 (default is 32768). STP places the highest-cost port on Admin-SW5 into DISCARDING state, breaking the loop.

```
# Admin-SW4 — force as root bridge
stp priority 4096
stp enable

# Admin-SW5 and Admin-SW6 — default priority
stp enable

# Verify: on Admin-SW5
display stp brief
# Expected: one port shows ALTE / DISCARDING state
```

STP Justification:

Physical redundancy between switches is essential so that a single cable failure does not isolate the hospital board network. However, redundant physical loops cause broadcast storms that would crash the network. STP provides the best of both worlds: redundant physical cabling for failover, with the loop logically broken until a failure is detected, at which point STP automatically unblocks the standby path within 30 seconds.

7.6 Eth-Trunk — Link Aggregation

Eth-Trunk 1 bonds two physical Ethernet links between Admin-SW4 and Admin-SW6 (Eth0/0/4 and Eth0/0/6 on both switches) into a single logical interface. The Eth-Trunk interface is configured as a trunk to carry both VLAN 10 and VLAN 20 traffic.

```
# Admin-SW4 Eth-Trunk configuration
interface Eth-Trunk 1
  port link-type trunk
  port trunk allow-pass vlan 10 20

interface Ethernet0/0/4
  eth-trunk 1
```

```
interface Ethernet0/0/6
  eth-trunk 1

# Admin-SW6 — identical configuration
interface Eth-Trunk 1
  port link-type trunk
  port trunk allow-pass vlan 10 20

interface Ethernet0/0/4
  eth-trunk 1

interface Ethernet0/0/6
  eth-trunk 1
```

Eth-Trunk Justification:

A single cable between the core administration switch and aggregation switch would be a bandwidth bottleneck for all VLAN 10 and VLAN 20 traffic combined. Eth-Trunk doubles the available bandwidth to 200 Mbps and provides link-level redundancy — if one physical cable is damaged or disconnected, traffic continues uninterrupted over the remaining link without any reconvergence delay.

7.7 Static Routing — Radiology Department

Three routers form a triangle: RadioGW-R5 connects to both RadioR8 and RadioR9. RadioR8 and RadioR9 are also connected to each other, providing a redundant path. Static routes are manually configured on each router with dual next-hops for every destination.

```
# RadioGW-R5 static routes
ip route-static 172.16.10.0 255.255.255.248 172.16.1.2
ip route-static 172.16.10.0 255.255.255.248 172.16.2.2
ip route-static 172.16.11.0 255.255.255.248 172.16.2.2
ip route-static 172.16.11.0 255.255.255.248 172.16.1.2
ip route-static 172.16.3.0  255.255.255.252 172.16.1.2

# RadioR8 static routes
ip route-static 172.16.11.0 255.255.255.248 172.16.3.2
ip route-static 172.16.11.0 255.255.255.248 172.16.1.1
ip route-static 0.0.0.0     0.0.0.0         172.16.1.1

# RadioR9 static routes
ip route-static 172.16.10.0 255.255.255.248 172.16.3.1
ip route-static 172.16.10.0 255.255.255.248 172.16.2.1
ip route-static 0.0.0.0     0.0.0.0         172.16.2.1
```

Static Routing Justification:

Radiology workstations transfer large DICOM imaging files where latency and path predictability are critical. Dynamic routing protocols can reroute traffic through unexpected paths during convergence events, which could interrupt an imaging session mid-transfer. Static routing ensures that imaging data always follows exactly the paths the administrator has defined, with no protocol overhead and no unexpected rerouting.

7.8 RIP v2 — Routing Information Protocol

RIP version 2 is configured on HospCore-R1 and all six departmental routers. Auto-summarization is disabled (undo summary) to allow /29 and /30 subnets to be advertised correctly without being collapsed into classful summaries. Each router advertises only its directly connected major networks.

```
# HospCore-R1 — advertises all backbone links
rip 1
  version 2
  undo summary
  network 10.0.0.0

# PatReg-R2
rip 1
  version 2
  undo summary
  network 10.0.0.0
  network 192.168.1.0

# EmergGW-R7
rip 1
  version 2
  undo summary
  network 10.0.0.0
  network 192.168.30.0

# (Same version 2 + undo summary pattern on all other routers)
```

RIP Justification:

RIP automatically distributes routing information between all departmental routers through HospCore-R1. Without RIP, every router would need a manual static route to every other department's subnet — 5 routes per router × 7 routers = 35 static routes to maintain. With RIP, adding a new department only requires enabling RIP on the new departmental router; all other routers learn its subnet automatically within one update interval (30 seconds).

7.9 ACL — Access Control List

ACL 2000 is a standard (basic) ACL configured on EmergGW-R7. It contains a single deny rule targeting the General Ward source subnet (192.168.30.8/29, expressed with wildcard mask 0.0.0.7). The ACL is applied outbound on the ICU sub-interface GE0/0/1.10, so any packet from the ward subnet that would exit toward ICU devices is dropped at the router.

```
# EmergGW-R7 ACL configuration
acl number 2000
  rule 5 deny source 192.168.30.8 0.0.0.7

interface GigabitEthernet0/0/1.10
  dot1q termination vid 10
  ip address 192.168.30.1 255.255.255.248
  arp broadcast enable
  traffic-filter outbound acl 2000
```

ACL Justification:

ICU patient monitoring equipment must never be accessible to general ward staff. If a ward PC were compromised, an attacker could not use it to probe or interfere with ICU monitoring systems because

all such traffic is dropped by the ACL at the router. The ACL is one-directional — ICU devices can still reach the ward network outbound, which is required for ICU staff to access shared hospital systems. This asymmetric enforcement is intentional and mirrors real-world hospital security policy.

7.10 FTP — File Transfer Protocol

FTPSrv-R6 acts as a dedicated medical records gateway router connecting three FTP server devices. Each server is attached on a separate /30 subnet via GE0/0/1 (Server1 — 1.1.1.x), GE0/0/2 (Server2 — 1.1.2.x), and GE0/0/3 (Server3 — 1.1.3.x). The FTP service runs on each server device directly via the eNSP Server GUI (Server Info → FtpServer → Start). The 1.0.0.0 network is advertised via RIP so all department routers can reach all three servers through HospCore-R1.

```
# FTPSrv-R6 FTP server configuration
ftp server enable

aaa
  local-user medrecords password cipher MedFTP@Hosp2026
  local-user medrecords privilege level 3
  local-user medrecords ftp-directory flash:
  local-user medrecords service-type ftp

# Connecting from any department router:
ftp 1.1.1.2
# User: medrecords
# Password: MedFTP@Hosp2026
# [ftp] prompt appears on successful login
```

FTP Justification:

Hospital departments need access to shared forms, discharge templates, and regulatory policy documents. Three dedicated FTP servers — Primary (active records), Backup (departmental copies), and Archive (discharged patients) — provide both redundancy and data segregation. If Server1 becomes unavailable, Server2 remains accessible for critical operations. The Archive server ensures long-term patient history is kept separate from active records, satisfying healthcare data retention requirements. FTPSrv-R6 acts as the single gateway for all three servers, simplifying routing — only one RIP-advertised network (1.0.0.0) is needed for the entire FTP infrastructure.

7.11 Telnet — Remote Router Management

Telnet is enabled on all six departmental routers. Each router's VTY line (0–4) is configured with a privilege level 15 password. HospCore-R1 can open a Telnet session to any router's backbone IP to remotely access its CLI and issue configuration commands.

```
# All departmental routers — identical Telnet config
telnet server enable

user-interface vty 0 4
  user privilege level 15
  set authentication password cipher HospNet@2026

# Example: R1 managing PatReg-R2 remotely
<HospCore-R1> telnet 10.0.1.2
# Enter password: HospNet@2026
<PatReg-R2>            ← now inside R2 CLI from R1
```

```
# All departmental router management IPs:
# PatReg-R2    10.0.1.2
# AdminGW-R3   10.0.2.2
# PharmaGW-R4  10.0.3.2
# RadioGW-R5   10.0.4.2
# FTPSrv-R6    10.0.5.2
# EmergGW-R7   10.0.6.2
```

Telnet Justification:

In a hospital, network routers are distributed across multiple floors and departments. Physically visiting each network closet to reconfigure a router after a topology change, security event, or software update is time-consuming and disruptive. Telnet from HospCore-R1 allows the core network administrator to manage any router in the building from a single console. This centralised management model reduces mean time to repair (MTTR) and allows out-of-hours changes without physical site visits.

8. Testing & Verification

Each technology is verified independently before running end-to-end tests. The table below documents the test command, expected output, and pass condition for every implemented technology.

Test #	Technology	Command / Action	Expected Output	Pass Condition
1	VLSM	display ip interface brief on PatReg-R2	GE0/0/1 = .1/29, GE0/0/2 = .9/29, GE0/0/3 = .17/29	Three /29 subnets from same /24
2	DHCP	Admit-PC1: ipconfig	IP 192.168.1.2–5, mask /29, gateway 192.168.1.1	IP auto-assigned in correct pool range
3	DHCP verify	PatReg-R2: display dhcp server statistics	Addresses allocated > 0 per pool	Allocation count matches online PCs
4	VLAN	Admin-SW4: display vlan	VLAN 10 and 20 active; access ports assigned correctly	Both VLANs exist; ports in correct VLAN
5	Router on Stick	Board-PC1 ping AdminStaff-PC1	Reply TTL=127 (one router hop)	Ping succeeds; TTL confirms router transit
6	STP	Admin-SW5: display stp brief	One port shows ALTE / DISCARDING	Loop broken; redundant port blocked
7	STP root	Admin-SW4: display stp	Line: This bridge is the root	SW4 elected as root bridge
8	Eth-Trunk	Admin-SW4: display eth-trunk 1	Both Eth0/0/4 and Eth0/0/6 show Selected	Both member ports active and aggregated
9	Static Route	RadioGW-R5: display ip routing-table	172.16.10.0/29 and 172.16.11.0/29 with proto Static	Routes present with Static protocol
10	Static Route	Radio-PC1 ping Radio-PC3	Reply received — cross-router path works	End-to-end ping via static triangle
11	RIP	HospCore-R1: display ip routing-table protocol rip	All dept subnets listed with proto RIP	All 6 dept subnets learned via RIP
12	RIP neighbors	HospCore-R1: display rip 1 neighbor	Six active neighbors (10.0.x.2)	All dept routers showing as RIP peers

Test #	Technology	Command / Action	Expected Output	Pass Condition
13	ACL deny	Ward-PC1 ping ICU-PC1	Request timeout — all packets dropped	ACL blocking ward-to-ICU traffic
14	ACL permit	ICU-PC1 ping Ward-PC1	Reply received from Ward-PC1	Reverse direction not blocked by ACL
15	ACL verify	EmergGW-R7: display acl 2000	rule 5 deny source 192.168.30.8 0.0.0.7	ACL rule present and applied
16	FTP	HospCore-R1: ftp 1.1.1.2, login medrecords	[ftp] prompt appears after login	FTP session established successfully
17	FTP multi-dept	PatReg-R2: ftp 1.1.1.2	[ftp] prompt from a different router	All routers can reach FTP via RIP
18	Telnet	HospCore-R1: telnet 10.0.1.2	<PatReg-R2> prompt appears	Remote CLI session opened from R1
19	Telnet all	Repeat for 10.0.2.2 – 10.0.6.2	Each router's CLI opens successfully	All 6 routers remotely accessible
20	End-to-end	Admit-PC1 ping 192.168.10.2, .20.2, .30.2, 172.16.10.2, 1.1.1.2	All 5 pings reply	Full cross-department connectivity confirmed

9. Design Assumptions and Limitations

- All routers used are basic routers but are assumed to be (Huawei AR2220) and all switches are Huawei S3700, consistent with device availability in the eNSP simulator.
- Telnet is used for remote management as it is natively supported in eNSP. In a production hospital deployment, SSH would replace Telnet to provide encrypted management sessions.
- RIP version 2 is used for backbone dynamic routing with undo summary enabled. In a production environment, OSPF would be preferred due to faster convergence and native VLSM support; however RIP v2 is used here to satisfy the CCP technology requirement.
- DHCP lease times are left at eNSP default values. In production, shorter lease times (4–8 hours) would be configured to ensure IP addresses are released promptly between hospital shifts.
- Standard (basic) ACL 2000 is used for the ICU protection rule, which filters by source address only. ACL rule is not properly working on basic router but works on (Huawei AR2220).
- All Telnet VTY passwords are set to HospNet@2026 uniformly for demonstration clarity. Production deployments would use unique, complex passwords per device stored in a privileged access management (PAM) system.
- FTPSrv-R6 connects three separate Server devices (Server1, Server2, Server3) on dedicated /30 subnets via GE0/0/1, GE0/0/2, and GE0/0/3 respectively. FTP services are enabled directly on each server device via the eNSP Server GUI (Server Info→FtpServer→Start). This multi-server design provides redundancy and data segregation: Server1 handles active patient records, Server2 stores departmental backups, and Server3 archives discharged patient history.
- Static routing in Radiology uses dual next-hop entries for each destination to provide path redundancy. If one link fails, the router automatically uses the alternate next-hop without any administrative intervention.
- The network assumes a single physical site (one hospital building). In a multi-site deployment, WAN links and a more scalable routing protocol such as OSPF or BGP would be required between sites.

10. Conclusion

The City Hospital network in Huawei eNSP implements all eleven CCP technologies — VLSM, DHCP, VLANs, Router-on-a-Stick, STP, Eth-Trunk, Static Routing, RIP v2, ACL, FTP, and Telnet — each mapped to real needs: DHCP automates IP allocation, VLANs segment networks, ACL secures ICU access, static routing stabilizes radiology transfers, RIP v2 enables dynamic connectivity, and Telnet supports centralized management.

A scalable hub-and-spoke design centered on HospCore-R1 allows new departments via AR2220 routers with /30 links and RIP, requiring no reconfiguration.

Testing confirms full operation: DHCP works, inter-VLAN routing via sub-interfaces, STP prevents loops, Eth-Trunk provides redundancy, static routes handle radiology traffic, RIP shares all subnets, ACL enforces ward/ICU policy, FTP is accessible, and Telnet enables full remote control.